

Worked Answer Key

Page 1: 1. $6 : 4 = 3 : 2$. 2. There are 10 animals, so dogs to all is $7 : 10$.

Page 2: 1. Yes: $2 \times 4 = 8$ and $3 \times 4 = 12$. 2. Multiply both parts by 3: $15 : 6$; blank **6**.

Page 3: 1. $12 \div 3 = 4$ groups; $2 \times 4 = 8$ cups flour. 2. $5 \times 4 = 20$, so $4 \times 4 = 16$; blank **16**.

Page 4: 1. $9 : 6 = 3 : 2$. 2. Total $8 + 12 = 20$; fiction fraction $8/20 = 2/5$, ratio fiction to all $8 : 20 = 2 : 5$.

Page 5: 1. Compare with the same number of red: $6 : 4 = 42 : 28$ and $9 : 7 = 36 : 28$. Since $42 > 36$, the first group has more blue per red. 2. Yes: $3 \times 4 = 12$ and $4 \times 4 = 16$.

Page 6: 1. $8 \div 2 = 4$ groups; $3 \times 4 = 12$ walking laps. 2. $12 \div 4 = 3$ groups; $3 \times 3 = 9$ lemons.

Page 7: 1. $18 \div 6 = \$3$ per notebook. 2. $240 \div 4 = 60$ miles per hour.

Page 8: 1. $5 \times 3 = 15$, $5 \times 7 = 35$, $5 \times 10 = 50$ pages. 2. $28 \div 4 = \$7$ each; $7 \times 9 = \$63$.

Page 9: 1. A: $6 \div 3 = \$2$ each; B: $12 \div 5 = \$2.40$ each. Store A is lower. 2. $18 \div 2 = 9$ minutes per mile; $9 \times 5 = 45$ minutes.

Page 10: 1. $12 \div 3 = 4$ liters per minute. 2. $45 \div 5 = 9$ pages per minute; $9 \times 8 = 72$ pages.

Page 11: 1. $35/100 = 0.35$. 2. $0.62 = 62/100 = 62\%$.

Page 12: 1. $48/100 = 48\%$. 2. $25\% = 1/4$; $80 \div 4 = 20$.

Page 13: 1. $10\% = 1/10$; $340 \div 10 = 34$. 2. $50\% = 1/2$; $126 \div 2 = 63$.

Page 14: 1. $75\% = 3/4$; $60 \div 4 = 15$, $15 \times 3 = 45$. 2. $20\% = 2 \times 10\%$; $25 + 25 = 50$.

Page 15: 1. $30/120 = 1/4 = 25\%$. 2. $40\% = 4/10$; if 28 is four tenths, one tenth is 7 and the whole is 70.

Page 16: 1. $25\% = 1/4$; $200 \div 4 = 50$ biographies. 2. 35% of 80 is 28; not tomatoes $80 - 28 = 52$.

Page 17: 1. 25% of 40 = \$10; sale price $40 - 10 = \$30$. 2. 10% of 60 = \$6; sale price $60 - 6 = \$54$.

Page 18: 1. 8% of $50 = \$4$; total $50 + 4 = \$54$. 2. 20% of $25 = \$5$; total $25 + 5 = \$30$.

Page 19: 1. Use a rate because each inch represents the same number of miles: $6 \times 4 = 24$ miles. 2. Use a percent because the question asks what part of all the games were wins, expressed per hundred: $18/24 = 3/4 = 75\%$.

Page 20: 1. $240 \div 12 = 20$ boxes; $25\% = 1/4$; $20 \div 4 = 5$ blue boxes. 2. $36 \div 3 = 12$ miles/hour; $12 \times 5 = 60$ miles; $60/100 = 60\%$.